# Java Full Stack Development — JavaScript Assessment

Student Examination Paper

Examination Details

Intermediate + Advanced

Duration

## 90 minutes

Total Questions

## 50

Total Marks

## 100

Question Types

### Mixed format

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Date: 22 . 10 .2026  Batch: 8

# Instructions

1. Answer all 50 questions.
2. Section A contains 25 MCQs, 1 mark each.
3. Section B contains 10 output-prediction questions, 2 marks each.
4. Section C contains 5 debugging questions, 3 marks each.
5. Section D contains 5 coding questions, 5 marks each.
6. Section E contains 5 practical and application-based questions, 5 marks each.
7. For coding questions, write valid JavaScript and show the expected output wherever applicable.

Correction to the marks distribution: Section D carries 3 marks per question and Section E carries 5 marks per question. The total is therefore exactly 100 marks.

# Section A — Multiple Choice Questions

25 questions × 1 mark = 25 marks

Choose the one correct answer for each question.

1. What is the output of the following code?

```
console.log(typeof NaN);
```

A. `"NaN"` B. `"number"` C. `"undefined"` D. `"object"`

Ans: B

2. Which declaration allows reassignment but does not allow redeclaration in the same scope?

A. `var` B. `const` C. `let` D. `static`

`Ans : C`

3. What is the output?

```
console.log("5" + 2);
console.log("5" - 2);
```

A. `7` and `3` B. `"52"` and `3` C. `"52"` and `"52"` D. `7` and `"3"`

**`Ans : B`**

4. Which expression checks both value and data type?

A. `a = b` B. `a == b` C. `a === b` D. `a != b`

**`Ans : C`**

5. What is the result of `Boolean("false")`?

A. `false` B. `true` C. `undefined` D. `NaN`

**`Ans : B`**

6. Which operator returns a value based on a condition in a single expression?

A. Spread operator B. Ternary operator C. Assignment operator D. Type operator

**`Ans : B`**

7. What is the output?

```
let marks = 75;

if (marks >= 90) {
    console.log("A");
} else if (marks >= 60) {
```

```
    console.log("B");
} else {
    console.log("C");
}
```

A. A B. B C. C D. No output

**Ans : B**

8. Which loop is designed to iterate over the values of an iterable such as an array?

A. `for...in` B. `for...of` C. `if...else` D. `switch`

**Ans : B**

9. What does `"JavaScript".slice(-6)` return?

A. `"Java"` B. `"Script"` C. `"JavaS"` D. `"ipt"`

**Ans : B**

10. Which method converts a string into an array using a delimiter?

A. `join()` B. `concat()` C. `split()` D. `splice()`

**Ans : A**

11. What is the output?

```
const greet = (name) => `Hello, ${name}`;
console.log(greet("Ragu"));
```

A. `Hello, name` B. `Hello, Ragu` C. `undefined` D. `Hello, ${name}`

**Ans : B**

12. What does a `return` statement do inside a function?

A. Prints a value automatically B. Sends a value back to the caller and ends that function execution C. Repeats the function D. Declares a variable

**Ans : B**

13. Which array method adds one or more elements to the end of an array and returns the new length?

A. `pop()` B. `shift()` C. `push()` D. `unshift()`

**Ans : C**

14. Which array method creates a new array containing elements that satisfy a condition?

A. `filter()` B. `forEach()` C. `reduce()` D. `find()`

**Ans : A**

15. What does `map()` normally return when called on an array?

A. A single number B. A new array containing callback results C. Only the first matching element D. A Boolean value

**Ans : B**

16. Which array method is most suitable for calculating the total of all numeric elements?

A. `some()` B. `find()` C. `reduce()` D. `every()`

**Ans : C**

17. What does `find()` return when no element matches its condition?

A. An empty array B. `false` C. `null` D. `undefined`

**Ans : D**

18. What does `every()` return when all array elements satisfy a condition?

A. The matching elements B. `true` C. The first matching element D. A new array

**Ans : B**

19. Which method checks whether at least one array element satisfies a condition?

A. `every()` B. `map()` C. `some()` D. `forEach()`

**Ans :C**

20. Which statement about `forEach()` is correct?

A. It creates a new array of callback results. B. It returns the first matching element. C. It is primarily used to execute a callback for each element. D. It automatically sorts the array.

**Ans : C**

21. What is the purpose of `Object.entries(obj)`?

A. Returns an array of the object's key-value pairs B. Returns only the object's values C. Deletes all object properties D. Converts the object into a JSON string

**Ans : A**

22. Which syntax accesses the `name` property of an object stored in `student`?

A. `student->name` B. `student.name` C. `student::name` D. `student#name`

**Ans : B**

23. Which method selects the first DOM element matching a CSS selector?

A. `document.getElementsByClassName()` B. `document.querySelector()` C. `document.createElement()` D. `document.write()`

**Ans : B**

24. What is the purpose of `event.preventDefault()` in a form's submit event handler?

A. It removes the form from the page. B. It prevents the browser's default form submission behavior. C. It automatically validates every input. D. It removes all event listeners.

**Ans : B**

25. Which event is generally appropriate for handling a form submission?

A. `mouseover` B. `keydown` C. `submit` D. `mouseenter`

**Ans : C**

# Section B — Output Prediction

10 questions × 2 marks = 20 marks

Write the exact output of each program.

26.

```
let x = 5;
console.log(x++ + ++x);
```

Output: 12

27.

```
console.log(typeof null);
console.log(typeof []);
```

Output: object object

28.

```
const text = "FullStack";
console.log(text.substring(4, 9));
```

Output: Stack

29.

```
function calculate(a, b = 3) {
    return a * b;
}

console.log(calculate(4));
console.log(calculate(4, 5));
```

Output:

12

20

30.

```
const nums = [2, 4, 6];
const result = nums.map(n => n / 2);
console.log(result);
```

Output: [1 , 2 , 3 ]

31.

```
const nums = [1, 2, 3, 4, 5];
console.log(nums.filter(n => n % 2 === 0));
```

Output: [ 2 , 4 ]

32.

```
const nums = [10, 20, 30];
const result = nums.reduce((total, n) => total + n, 0);
console.log(result);
```

Output: 60

33.

```
const students = [
    { name: "Arun", mark: 65 },
    { name: "Bala", mark: 85 },
    { name: "Cathy", mark: 45 }
];

console.log(students.find(s => s.mark >= 60).name);
```

Output: Arun

34.

```
const values = [3, 10, 2];
values.sort();
console.log(values);
```

Output: [ 2 , 3 , 10 ]

35.

```
const user = {
    name: "Ravi",
    age: 25
};

const { name, age } = user;
console.log(`${name}-${age}`);
```

Output: Ravi-25

# Section C — Debugging

5 questions × 3 marks = 15 marks

Identify the error and rewrite the code correctly.

36. Incorrect comparison

```
let age = 18;

if (age = 18) {
    console.log("Eligible");
}
```

Task: Correct the condition so it checks whether `age` equals 18 without assigning a new value.

**Corrected code :**
let age =18;

if (age == 18){

console.log(“Eligible”)

}

37. Incorrect string method

```
const message = "javascript";
console.log(message.toUppercase());
```

Task: Correct the code to print `JAVASCRIPT`.

**Corrected code :**

const message = "javascript";

console.log(message.toUpperCase());

38. Incorrect array filtering

```
const numbers = [10, 15, 20, 25, 30];

const result = numbers.filter(n => {
    n > 20;
});

console.log(result);
```

Task: Correct the callback so the result contains all numbers greater than 20.

**Corrected Code :**

```
const numbers = [10, 15, 20, 25, 30];

const result = numbers.filter(n => {
    return n > 20;
});

console.log(result);
```

39. Incorrect object access

```
const employee = {
    name: "Kumar",
    salary: 30000
};

console.log(employee[department]);
```

Task: Print the employee's salary using the correct property access.

**Corrected Code :**

```
const employee = {
    name: "Kumar",
    salary: 30000
};

console.log(employee[“salary”]);
```

40. Incorrect form handling

```
const form = document.getElementById("studentForm");
```

```
form.addEventListener("submit", function(event) {
    const name = document.getElementById("studentName").value;
    console.log(name);
});
```

Task: Update the event handler to prevent the browser's default form submission before processing the input.

**Corrected Code :**

```
const form = document.getElementById("studentForm");

form.addEventListener("submit", function(event) {
        event.preventDefault()

   const name = document.getElementById("studentName").value;
   console.log(name);
});
```

# Section D — Coding Questions

5 questions × 3 marks = 15 marks

Write a JavaScript program for each task. Use appropriate variable names and correct syntax.

41. Number Classification

Write a function `classifyNumber(num)` that returns:

- `"Positive"` if the number is greater than zero.
- `"Negative"` if the number is less than zero.
- `"Zero"` if the number equals zero.

**Code :**

```
let a = 0

function classifyNumber(num){

   if(num > 0){

      console.log("Positive")

   }

   else if(num < 0){

      console.log("Negative")
```

```
    }
    else{
        console.log("Zero")
    }
}
classifyNumber(a)
```

42. String Processing

Write a function `countVowels(str)` that returns the number of vowels (`a`, `e`, `i`, `o`, `u`) in a string, ignoring letter case.

Example: `countVowels("JavaScript")` should return `3`.

**Code :**

```
function countVowels(str) {
    let count = 0;
    for (let ch of str.toLowerCase()) {
        if ("aeiou".includes(ch)) {
            count++;
        }
    }
    return count;
}
console.log(countVowels("JavaScript"));
```

43. Array Transformation

Given an array of numbers, use `map()` to create a new array containing the square of each number.

Example input: `[2, 3, 4, 5]`

Expected output: `[4, 9, 16, 25]`

**Code:**

```
let arr = [2 ,3 , 4, 5]

let new_arr = arr.map((val) => val*val)

console.log(new_arr)
```

44. Filter and Sort

Given an array of numbers, use `filter()` to retain only numbers greater than 10 and then sort them in ascending numeric order.

Example input: `[25, 5, 12, 40, 8, 15]`

Expected output: `[12, 15, 25, 40]`

**Code:**

```
let arr = [25 , 5 ,12 , 40 ,8 ,15]

let new_arr = arr.filter((val)=> val>10).sort((a,b) => a-b)

console.log(new_arr)
```

45. Object Method

Create an object named `product` with properties `name`, `price`, and `quantity`. Add a method named `getTotal()` that returns the total price based on the price per item multiplied by quantity.

Example: If `price` is `250` and `quantity` is `4`, the method should return `1000`.

**Code** :

```
const product = {
        name: "Laptop",
        price: 250,
        quantity: 4
    };
    product.getTotal = function () {
```

```
    return product.price * product.quantity;
};

console.log(product.getTotal());
```

# Section E — Practical and Application-Based Questions

5 questions × 5 marks = 25 marks

46. Student Marks Analyzer

Create a function `analyzeMarks(marks)` that accepts an array of student marks and returns an object containing:

- `total`: sum of all marks.
- `average`: average mark.
- `passed`: number of students who scored at least 35.

Use `reduce()` to calculate the total and `filter()` to count the students who passed.

Example input:

```
[80, 30, 65, 90, 20]
```

Expected result:

```
{
    total: 285,
    average: 57,
    passed: 3
}
```

**Code :**

```
let marks = [80, 30, 65, 90, 20]
function analyzeMarks(marks) {
    let total = marks.reduce((sum, mark) => {
        return sum + mark;
    }, 0);
    let passed = marks.filter(mark => mark >= 35);
    return {
        total: total,
        average: total / marks.length,
        passed: passed.length
    };
}

console.log(analyzeMarks(marks));
```

47. Employee Data Processing

Given the following array:

```
const employees = [
    { name: "Arun", salary: 25000, department: "IT" },
    { name: "Bala", salary: 40000, department: "HR" },
    { name: "Cathy", salary: 50000, department: "IT" },
    { name: "Divya", salary: 35000, department: "IT" }
];
```

Write JavaScript code to:

1. Select only employees in the IT department.
2. Extract their names using `map()`.
3. Calculate the total salary of all IT employees using `reduce()`.

Expected results:

- Names: `["Arun", "Cathy", "Divya"]`
- Total IT salary: `110000`

**Code :**

```
const employees = [
   { name: "Arun", salary: 25000, department: "IT" },
   { name: "Bala", salary: 40000, department: "HR" },
   { name: "Cathy", salary: 50000, department: "IT" },
   { name: "Divya", salary: 35000, department: "IT" }
];
let itEmployees = employees.filter(emp => emp.department == "IT");
let names = itEmployees.map(emp => emp.name);
let totalSalary = itEmployees.reduce((total, emp) => {
   return total + emp.salary;
}, 0);
console.log("Names:", names);
console.log("Total IT salary:", totalSalary);
```

48. Form Event Handling

Create an HTML form with a name input and a submit button. Write JavaScript that:

1. Selects the form using `getElementById()`.
2. Registers a `submit` event listener.
3. Prevents the default form submission.
4. Reads the name entered by the user.
5. Displays a welcome message on the page.

**Code :**

```
<!DOCTYPE html>

<html>

<head>

   <title>Welcome Form</title>

</head>

<body>

   <form id="studentForm">

      <input type="text" id="studentName" placeholder="Enter your name">

      <button type="submit">Submit</button>

   </form>

   <h2 id="message"></h2>

   <script>

      let form = document.getElementById("studentForm");

      form.addEventListener("submit", function(event) {

         event.preventDefault();

         let name = document.getElementById("studentName").value;

         document.getElementById("message").innerHTML =

            "Welcome, " + name + "!";

      });

   </script>

</body>
```

</html>

49. Array Methods — Logic Challenge

Given:

```
const numbers = [12, 5, 18, 7, 24, 9, 30];
```

Write JavaScript code to:

1. Use `filter()` to select numbers greater than 10.
2. Use `map()` to multiply each selected number by 2.
3. Use `reduce()` to calculate the sum of the transformed values.
4. Use `some()` to check whether any original number is greater than 25.
5. Use `every()` to check whether all original numbers are positive.

**Code :**

```
const numbers = [12, 5, 18, 7, 24, 9, 30];

let greater_10 = numbers.filter((val) => val >10)

console.log(greater_10)

let mul_2 = greater_10.map((val) => val*2)

console.log(mul_2)

let trans = mul_2.reduce((tot , val) => tot + val , 0)

console.log(trans)

let great_25 = numbers.some((val) => val>25)

console.log(great_25)

let positive = numbers.every((val) => val>0)

console.log(positive)
```

50. Mini Project — Student Registration Form

Build a simple student registration form using HTML and JavaScript with these fields:

- Student name
- Email
- Age

Requirements:

1. Handle submission using `addEventListener("submit", ...)`.
2. Use `event.preventDefault()` to prevent page reload.
3. Validate that the name is not empty and age is at least 18.
4. Create a student object containing the entered values.
5. Display the registered student's details on the page using DOM manipulation.

**Code:**

```
<!DOCTYPE html>

<html>

<head>

   <title>Welcome Form</title>

</head>

<body>

   <form id="studentForm">

      <label for="studentName">StudentName</label>

      <input type="text" id="studentName" >

      <label for="email">Email</label>

      <input type="email" id="email">

      <label for="age">Age</label>

      <input type="age"  id="age">

      <button type="submit">Submit</button>

   </form>

   <h2 id="message"></h2>

   <script>

      let form = document.getElementById("studentForm");
```

```
let msg = document.getElementById("message")
form.addEventListener("submit", function(event) {
    event.preventDefault();
    let Name = document.getElementById("studentName").value;
    let Age = Number(document.getElementById("age").value);
    let Email = document.getElementById("email").value;
    console.log(Age)
    if(Name == ""){
        msg.innerHTML = "Please enter student name.";
        return
    }
    if(Age <= 18){
        msg.innerHTML = "Age atlest 18"
        return
    }
    let student ={
        studentname :Name,
        Age :Age,
        Email : Email
    }
    msg.innerHTML = `
    <h3>Registered Student Details</h3><br>
    <p>Name : ${student.studentname}</p><br>
    <p>Age : ${student.Age}</p><br>
    <p>Email : ${student.Email}</p>
```

```
            `

        });
    </script>


</body>
</html>
```